Installation Guide
PSS®E 35.2.0

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SIEMENS

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Chapter 1
Requirements

1.1. Hardware, Operating System, and Privileges

The following are offered as baseline suggestions. Faster is always better, and a Solid State Drive, SSD, rather than a rotating drive Hard Drive, is best.

- Intel® 4th generation Haswell chip or later. The Haswell generation of chips was released in May 2014.
- 16 GB or more of RAM is highly recommended. PSS®E-35 is now a 64-bit application and can take advantage of additional RAM.
- Approximately 1 GB of free disk space is required for a full installation of PSS®E. Be sure to allow sufficient room for working files. You will need to allow additional disk space to support Windows® use of virtual memory.
- A discrete graphics card is optimal. We have seen performance issues with some of the integrated graphic cards/chips.
- Microsoft Windows 7 x64®, Microsoft Windows 10 x64®, Microsoft Windows® Server 2016 x64 or Microsoft Windows® Server 2016 x64. PSS®E-35 is a 64-bit application and will not run on a 32-bit OS.
- Administrative privileges are required to install PSS®E, but not to run the program. You must have administrative privileges on the target machine in order to install PSS®E. Setup will check for your privileges and will stop the installation process with a warning if you do not have the necessary privileges.

1.2. Display Resolution

PSS®E is written for a minimum display resolution of 1024x768 pixels. We highly recommend using 1024x768 resolution with the "small fonts" option enabled, and using a 17" or larger monitor. In general, running with less than 1024x768 resolution will yield suboptimal results.

An AGP card with color palette at a minimum setting of 32 bit (True Color) is highly recommended to improve graphics performance.

1.3. Memory

PSS®E requires a significant amount of computer memory in order to execute. This memory may be either physical RAM or virtual memory in the form of a swap file on your hard disk. For most efficient operation, however, your system should have sufficient physical memory to contain the PSS®E program plus those data arrays that are dynamically allocated while the program is running. Because of these dynamic arrays, it is hard to determine the exact amount of memory that may be required in a particular case. However, following are some approximate memory requirements that can be used for planning purposes:

Memory	System	Application
65MB	12,000 bus	power flow

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Memory	System	Application
75MB	12,000 bus	dynamics
95MB	50,000 bus	power flow
110MB	50,000 bus	dynamics
135MB	100,000 bus	power flow
160MB	100,000 bus	dynamics
175MB	200,000 bus	power flow
210MB	200,000 bus	dynamics

After PSS®E is installed, you can use the appropriate system monitoring tool, Task Manager, to determine the exact amount of memory that is being consumed in any given case.(You should also verify that there is sufficient physical memory on your system to keep PSS®E from swapping to disk.)

Keep in mind that the operating system itself, along with any other programs that may be running, can consume a great deal of memory. Here again, a monitoring tool such as Task Manager can be very helpful in determining the amount of memory that is really available for PSS®E. (Also be aware that running multiple instances of any PSS®E program will increase memory demands.)

It is possible to run PSS®E even if there is not sufficient physical memory on your system. Performance is sure to suffer in that case, but the program should still execute if enough virtual memory is available. Consult with your computer support people if you need to increase the amount of virtual memory on your system.

1.4. Hardware Lock/Software License

Unless your work group has a PSS®E network license, each PC running PSS®E must be equipped with a hardware lock (commonly called a dongle), which is shipped as part of the PSS®E installation package or a software license. Depending upon your installation requirements, the lock may connect to the USB interface.

USB lock: Do not connect the lock until after PSS®E has been fully installed.

Make sure that appropriate lock has been attached or a software license collected before attempting to run PSS®E.

See the PTILicenseManual.pdf for more information.

1.5. Network License

If your work group has a CodeMeter network license for PSS®E 35, a locally installed license is not required. Install PSS®E 35 and then refer to the PTILicenseManual.pdf for instructions to configure your client license setup.

1.6. Python

PSS®E includes an embedded Python interpreter. Python is an open-source scripting language (http://www.python.org). PSS®E also includes an extension module for Python that allows Python programs to access the PSS®E API. The libraries for the Python interpreter are required to install PSS®E. Some of the Python scripts provided with PSS®E installation require Python for Windows extensions (pywin32) and wxPython modules. Instructions for the installation of Python are included in the PSS®E installer program. Python, Pywin 32 and WXPython are included in the PSS®E 35 Complete installer.

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1.7. CPU

PSS®sE-35 contingency and GIC analysis APIs have been optimized to take advantages of multiple processors. Significant performance gains can be achieved in contingency and GIC analysis by using the multiple processor enabled APIs.

PSS®E is now built as a 64-bit application; It can only run on 64-bit operating systems.

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Chapter 2
PSS®E Compilers

2.1. Overview

If you plan to use certain advanced modules for PSS®E program options, you will need to obtain the Intel Visual Fortran compiler, Version 19 for each system. The following list provides the combinations requiring the compiler:

- *Dynamics* (and you plan to compile and link the user written models or connection subroutines using CLOAD4.BAT or the "createusrdll" facility).
- *IPLAN* (and you plan to create a custom routine using CLIPLU.BAT).
- *Power flow* (and you want to add a user activity using CLPSSUSR.BAT).

The Intel Visual Fortran compiler is available in several configurations. Please consult the PSS®E website for guidance in choosing the compiler configuration that is right for you.

PSS®E will run without the Intel Visual Fortran compiler, but you will not be able to make system-wide modifications of PSS®E operation without it.

If you built any of the advanced modules noted above using a version of the Microsoft/Digital/COMPAQ compiler, you will need to recompile and relink them using the Intel compiler.

In case of difficulties, check to be certain that any batch file you are executing was supplied with PSS®E-35. or, in the case of compile files, was generated by PSS®E 35 activity DYRE.

2.2. Compiler Installation

User-written software included with PSS®E (e.g. Conec, Conet, user-written dynamics models, programs that call USRCAS) should be compiled with the version of the compiler used to create that version of the PSS®E product. We cannot provide support for problems that arise when mixing code from different compilers.

If you need to compile files for use with older versions you will need to be able to correctly install and access the Fortran compiler, possibly the C compiler, and also the linker associated with the particular version of PSS®E. PSS®E 35 was built with IVF 19.0. Any version of IVF 15.0 or later can be used to build extension libraries (for example dsamusr.dll).

If the IVF compiler that you use and the one used to build PSS®E 35 are different, it is required that the later run-time subroutines be used.

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Chapter 3
Installation

The following sections assume that a new installation is being performed. If a previous installation of PSS®E-35 is detected, a maintenance script will be started to allow you to remove the previous installation or install another copy of minor version of PSS®E 35. See also Section 4.2, Manually Uninstalling PSS®E.

3.1. Support for Multiple Versions of PSS®E

PSS®E now supports installations of multiple Minor versions of the program at once. You no longer need to uninstall PSS®E-35.0.0, for example, to install PSS®E-35.1.0. This allows you to "test" new versions of the program before you move off the version you are currently using for production. Update releases, e.g. 35.0.1 cannot be installed separately and will need to replace the existing 35.0.0 installation.

3.2. Silent Installation

You can install PSS®E-35 silently by following these guidelines. Note that the prerequisite installers are not included in this process (i.e. you cannot use the silent option to install the "Complete" installation of PSSE). This silent installation directive only applies to the PSS®E application.

Once you perform the "Complete" installation, the PSSE installation located at `C:\Program Files\PTI\PSSE35 Installer Files\35.x\PSSE` can be used to create a silent installation script. Each of the executables located at `C:\Program Files\PTI\PSSE35 Installer Files\35.x` will need to be installed separately. Not all of these executables can be installed silently and may follow a different method for setting up a silent installation.

/r : record mode

In order to install PSS®E in silent mode, you must first run Setup.exe with the /r option to generate a response file. Running setup.exe with the command Setup.exe /r will store data entered during the installation in a file called Setup.iss. Setup.iss is created inside the system's Windows folder. You can provide an alternative response file name and location by using the /f1 option. Example (Setup /r /f1"C:\pssetemp\PSSE_silent.iss")

/s : silent mode

The command Setup.exe /s runs the installation in silent mode. By default, the file Setup.iss will be used to provide responses contained in a response file. Setup.iss must be located in the same directory as Setup.exe. (Response files are created by running Setup.exe with the /r option.) You can provide an alternative file name or location of the response file, use the /f1 option.

/f1 : specify response file name and path

The /f1 option allows you to specify where the response file is created and its name.

Example (Setup /s /f1"C:\pssetemp\PSSE_silent.iss").

The /f1 option is available when creating a response file (with the /r option) and when using a response file (with the /s option).

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/f2 : specify log file name and path

The /f2 option allows you to specify the name and location where the log file is created.

Example (Setup /s /f1"C:\pssetemp\PSSE_silent.iss" /f2" C:\pssetemp\setup.log")

To silently uninstall, you must first create a uninstall response file. This will uninstall PSS®E-35. You can then use this response file to uninstall other installations of PSS®E-35

Example (Setup /s /f1"C:\pssetemp\PSSE_silent_uninstall.iss").

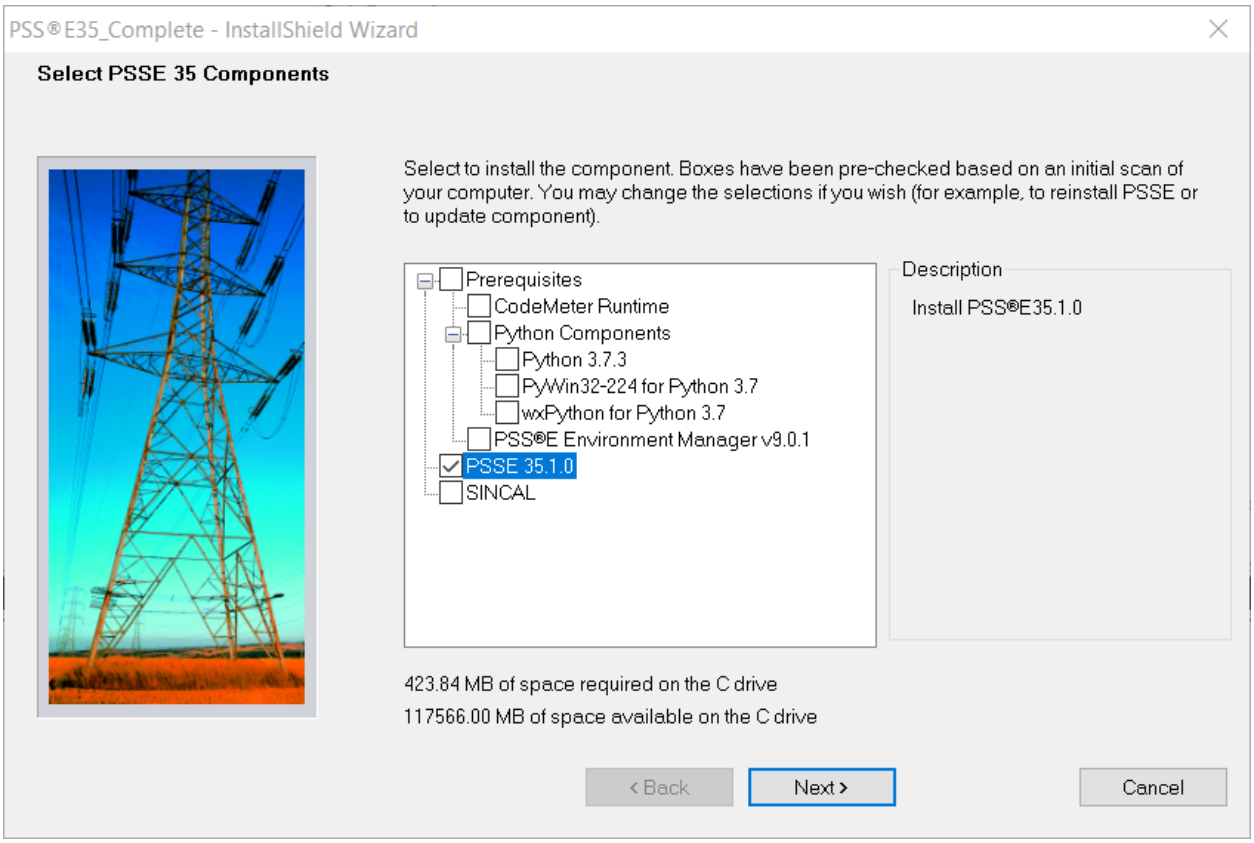
Then use the command:

Setup /s /f1"C:\pssetemp\PSSE_silent_uninstall.iss"

3.3. Complete Installation

PSS®E-35 can be installed when you run the PSS®E-35 Complete installer. The Complete installer will install Python, PyWin32, WxPython, CodeMeter runtime, SINCAL, the PSS®E Environment Manager and PSS®E. If you receive errors during installation, temporarily disable your anti-virus software and try installing again

If an installer for SINCAL is found, you will be given the option to launch the SINCAL installer.



Selected programs will be installed or updated before PSS®E is installed or updated. SINCAL will be installed last.

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The Complete installer should be downloaded from the following PSS®E webpage:https://siemens-pss.com

3.4. Preliminary Setup Screens

1. Review *ReadMe*

   The latest ReadMe Notes for PSS®E-35 will display. Read these notes very carefully, because they can contain critical information that you should consider before installing PSS®E-35 on your system.

2. Review *License Agreement*

   Siemens PTI requires that you accept the terms of our license agreement.

Note that this release of PSS®E has not been tested with versions of the IVF runtime libraries later than 19.0.

PSS®E keeps the Intel files in the PSSBIN folder. However, there is a potential hazard: any user-generated code compiled with version later than 19.0 may not be compatible with the version 19.0 libraries being installed.

3.5. Destination Directory

Specify a destination directory for PSS®E. The default directory is *C:\Program Files\PTI\PSSE35\35.x*, which should be satisfactory in most cases.

3.6. Program Folder

Specify the Program Folder (i.e., the name of the folder under **`Start>Programs>PSSE 35`**) that will be used to contain all the PSS®E-34 programs installed on the system. Type a Program Folder name or highlight an existing folder.

3.7. Operating Frequency / Choice Confirmation

Select the appropriate operating frequency for your studies. By default, PSS®E is set for 60 Hz, but you may also select 50 Hz operation. If you are unsatisfied with any of your installation choices, you may also use the Back button to move to the appropriate dialog and change your selections. Clicking Next will display an installation progress screen.

3.8. PSS®E-35 Command Prompt

The PSS®E-35 Command Prompt should be used any time a PSS®E-35 command such as, CLOAD4, CLOAD4NEW, PSSE, etc., needs to be run from the command line. Select the location for the shortcut PSS®E-35 Command Prompt, in the Start Menu or on your desktop.

3.9. Post PSS®E-Installation

If desired, review the Release Notes. At the end of the PSS®E installation, do not choose to restart the computer if the SINCAL installation was chosen. If you do restart, you will need to launch the SINCAL installer yourself with the file ..\SincalIns_15.0\Sincal\setup.exe.

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3.10. Example Files

When PSS®E-35 is **first** installed on a system, the contents of the current installation EXAMPLE directory are copied to the users "Documents" directory like this:

```
C:\Users\[username]\Documents\PTI\PSSE35\EXAMPLE
```

This directory is **NOT** updated with subsequent installations of PSS®E-35.

Note that your "Documents" folder may be located in a different directory, check with your IT department if you are unsure.

To update the user EXAMPLE directory with EXAMPLE files from the **latest** PSS®E-35 installation, start PSSE and then select File > Reset Example Directory from the menu. This will update the user EXAMPLE directory with the current contents of the installation EXAMPLE directory i.e. the latest set of EXAMPLE files.

3.11. SINCAL Installation

If the SINCAL installation is launched from the PSS®E-35 Complete installer, it will install the default options needed for use with PSS®E. If you launch the SINCAL installer yourself, you can choose the options such as Destination location and program options. Make sure to install the options (PSS SINCAL and PSS NETOMAC). Run setup.exe from your download location ..\SincalIns_15.0\Sincal\setup

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Chapter 4
Uninstalling

4.1. Uninstalling PSS®E-35

4.1.1. Backing Up Files

NOTE: Before uninstalling PSS®E-35, make sure you have saved any files in the PSS®E-35 installation directory that you do not want deleted.

If multiple releases of PSS®E-35 are installed, files for each release being uninstalled should be backed up. If only one release of PSS®E-35 is being uninstalled, other installation folders in the installation directory will not be modified by the installer program.

4.1.2. Running the Uninstaller

PSS®E-35 can be uninstalled in one of two ways:

1. Going to *Start>Settings>Control Panel>Add/Remove Programs*, selecting PSSE35.x
2. Attempting to reinstall a copy of PSS®E-35 minor version.

This next screen will ask you to confirm that you *really* want to uninstall the application and all features.

A status screen will show the progress of the uninstallation.

At this point, all the unmodified PSS®E-35 files will be deleted.

After you click **`Finish`** to exit from the uninstall program, any files which you added or modified AFTER the installation of PSS®E-35 will still be present in your PSS®E-35 installation directory. Therefore, if you want to completely remove PSS®E-35 you will probably need to manually delete the PSS®E-35 installation directory and any files which still remain there.

4.2. Uninstalling Python

The installer program for PSS®E-35 will not automatically uninstall supported versions of Python from the machine during the uninstallation process in order to preserve changes made outside of PSS®E-35

NOTE: Uninstalling Python may delete files that have been modified from their default state and/or remove files that have been downloaded separately to the Python installation folder (including external third-party modules that are not associated with PSS®E-35). Ensure that no other applications depend on Python and these files before uninstallation and that the respective files are backed up prior to continuing.

To uninstall Python versions supported by PSS®E-35, use the Add/Remove Programs options from Control Panel and follow the instructions in the following dialogs that appear from the Python installation program.

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4.3. Uninstalling pywin32 for Python

The installer program for PSS®E-35 will not automatically uninstall the pywin32 Python module from the machine during the uninstallation process due to possible external dependencies.

NOTE: There may be several instances of pywin32 listed; take note which version of Python that the pywin32 module belongs to before uninstalling and select accordingly.

To uninstall the pywin32 Python module, use the Add/Remove Programs options from Control Panel and select the version of pywin32 to be removed. Follow instructions in the dialog/window that appears from the installation program.

4.4. Uninstalling wxPython for Python

The installer program for PSS®E-35 will not automatically uninstall the wxPython Python module from the machine during the uninstallation process due to possible external dependencies.

The following are two ways to remove the wxPython Python module:

1. **Deleting the wxPython directory under Python's Lib/site-packages folder**

   Deleting the wxPython dist folder at C:\[Python-install-directory]\Lib\site-packages will remove the module from disk. The installation directory will vary from machine to machine if default options were not used during the initial install of Python.

2. **Running the pip Python module to uninstall wxPython**

   i. Open a Command Prompt as Administrator and change directories (cd) to your Python's installation directory with:

   ```
   cd C:\[Python-install-directory]\scripts
   ```

   An example would look similar to:

   ```
   cd C:\Python37x64\scripts
   ```

   ii. Run the pip uninstall command:

   ```
   pip uninstall wxpython
   ```

   Allow the uninstallation process to complete. Once complete, the Command Prompt window can be closed.

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Chapter 5
Support

5.1. PSS®E Manuals

The *PSS®E Program Operation Manual (POM)*, the *PSS®E Program Application Guide (PAG)*, the PSS®E GUI Users Guide, and the PSS®E Command Line Interface (CLI) Users Guide provide detailed information on how to work with PSS®E.

5.2. World Wide Web

If you have access to a web browser, you can check our web site for the most up to date information and FAQs. Just point your browser to www.siemens.com/power-technologies. Besides information on PSS®E, our web site contains information on the whole range of PTI products and services.

Please visit the Siemens PTI Web Site at www.siemens.com/power-technologies and the PSS®E User Software Delivery Platform https://siemens-pss.com for more information and product downloads.



Please note that access to the latest releases of PSS®E on the PSS® Software Delivery platform and to technical PSS®E support are only available to those users whose companies have purchased or renewed their PSS®E maintenance and support agreement for the current calendar year.

5.3. Technical Support

For fastest response to any questions, please send your questions/comments to:

https://siemens-pss.freshdesk.com/support/login

Alternatively, you can use the "Request for Technical Support" fax transmittal sheet on the following page. For telephone support between the hours of 8:00 a.m. and 5:00 p.m. (Eastern Standard Time) Monday through Friday, address your calls to the PSS®E support staff at 518-395-5075.

5.4. Reporting a Problem or Requesting a Feature

Whenever you report a problem with PSS®E, please provide as thorough a description as possible. Use the form included on the next page to help you provide the necessary information.

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5.4.1. Request for Technical Support

PSS®E Support
Siemens Power Transmission & Distribution, Inc. Power Technologies International Fax Number: 518-346-2777   For a prompt reply to your technical support questions, complete this form and fax it along with any additional information to Siemens PTI.
Name:
Company:
Phone:
Email Address:
PSS®E Version:
Date:
PC Manufacturer/Model:
CPU Type:
Operating System/Version:
Amount of RAM Installed:
Describe your problem as completely as possible, including all steps needed to replicate the problem:

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5.4.2. Submitting Bug Reports and Feature Requests

Bug reports should be submitted directly to PSS®E support at:

https://siemens-pss.freshdesk.com/support/login.

Please provide as much detail as possible. If submitting a report for a potential bug please include the steps taken, along with pertinent data files and scripts so that we may accurately reproduce the problem. If an issue does turn out to be a program bug, one of the following priorities will be assigned to it:

- High priority is given to issues that cause the program to crash or produce incorrect results with no published work around.
- Medium priority is given to issues that cause incorrect functionality, however the problem can be remedied by a published work around.
- Low priority is given to those issues that do not fall into the above categories, or are purely cosmetic in nature.

Every effort is made to address high and medium issues in a PSS®E point release. Some low priority items may also be include in a point release although this cannot be guaranteed.

Feature requests should be submitted directly to the PSS® Ideas Portal at:

http://siemens.com/pss-ideas

The Siemens PSS® Ideas Portal provides the PSS® community an easy and transparent way of having a voice into the product development roadmap. The entire community can view, vote and comment on Ideas created by any user. If your request already exists in the portal from another user, please vote on the existing idea rather than creating a new one. If you request doesn't already exist, please create a new one and provide your detailed user requirement. If you have more than one request, please create separate Ideas rather than group them all into one. Popular Ideas will be given priority in the features roadmap. Additionally, developers may provide direct feedback by commenting on Ideas and updating their development status. If you don't already have an account, please request access via support.

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